

Experimental Contact II: The Composite Gate

Entanglement from a Shared Drive and Exact Tsirelson Saturation in Finite Ring Cosmology

Yosef Akhtman

12 June 2026

Abstract

This note constructs the composite layer of the FRC quantum formalism — the gate through which every decisive experimental confrontation of the programme passes — and executes the first computation behind it. The construction adds nothing to the framework: composite ledger states are functions on the product of phase cycles, and the carrier constraint is supplied by the substrate itself, because the single global drive acts *diagonally* on a pair of subsystems and therefore conserves their relative phase offset. The synchronised drive orbits are the Bell states; entanglement is orbit coherence; channel conditioning by a core- C_8 Subject reduces a C_{16} Object to a winding doublet that is an exact qubit; the half-turn identity $g_O^8 = -1$ makes the forced two-outcome readout an exact σ_x measurement; and analyser settings are relational repositionings of Subject against Object along the carrier, with phase granularity $\pi/40$ in the configuration $C_{641}/S_{41}/O_{17}^{\times 2}$. In that configuration the correlators obey the singlet law $E(\Delta) = \cos(\pi\Delta/40)$ exactly, and the CHSH functional reaches $S = 2(\zeta_8 + \zeta_8^{-1}) = 2\sqrt{2}$ — Tsirelson saturation as a cyclotomic ring identity, $S^2 = 8$ — at carrier-native settings, while an exhaustive sweep over all 80^3 setting triples never exceeds it. No-signalling holds exactly, and the ledger-to-carrier reduction commutes at the composite level. The framework thus lands exactly on the quantum boundary: strictly above every local and epistemic-restriction model, strictly below every superquantum box, in agreement with loophole-free Bell experiments. All claims are machine-verified by exact arithmetic (`composite.py`; twelve checks, all passing).

1 The composite gate

The companion manuscript [1] establishes the single-system formalism — wave functions as windings, amplitudes as Subject-relative projections, Born weights as coincidence counts in the ledger — and lists the composite rule as its first open problem: in standard quantum mechanics independent systems compose by tensor product, while the join of two cycles in a carrier has order lcm, not the product. The resolution is that the two constructions answer different questions, and the framework needs both.

Definition 1.1 (Composite ledger). For Objects O_A, O_B with phase cycles Φ_A, Φ_B in a common Carrier, the composite ledger space is $\Lambda^{\Phi_A \times \Phi_B}$: ledger amplitudes assigned to *joint* phase configurations. Its rank is $n_A n_B$ — dimensions multiply, as the Subject’s bookkeeping of joint tallies requires. Local operations act on single factors; the pair-count Born rule of [1] extends verbatim.

What the carrier adds is not a smaller state space but a dynamical constraint, and it is forced rather than chosen:

Proposition 1.2 (One drive, conserved offset). *The carrier supplies a single drive, which advances every subsystem simultaneously: on joint configurations it acts diagonally, $(u, v) \mapsto (g_A u, g_B v)$. For a pair of equal cycles the relative offset uv^{-1} is a constant of motion; the joint configuration space foliates into drive orbits, and no carrier evolution connects distinct offsets. Verified on all 256 joint configurations of the C_{16} pair in $\mathbb{F}_{641}$; the synchronised orbit through $(1, 1)$ has length exactly 16 (`composite.py`).*

Definition 1.3 (Bell orbit states). A *synchronised orbit state* is the coherent ledger superposition of one drive orbit, $\Psi = \sum_j |g_A^j u_0\rangle \otimes |g_B^j v_0\rangle$. In the winding representation, $\Psi = \sum_s |s\rangle \otimes |-s\rangle$ up to offset phases: the maximally correlated state of the pair.

Entanglement is therefore *orbit coherence*: a synchronised orbit state assigns one amplitude to each joint configuration of a single orbit and cannot be factored into independent single-cycle states. The carrier does not forbid product states; it makes correlation the default, because any two symmetry-complete subsystems already share at least the quarter-turn core and one drive. This localises, within FRC, exactly the independence assumption whose status the real-versus-complex dispute has isolated [10, 11]: in a finite relational substrate, nothing is ever fully independent, and “independent sources” is an approximation whose quality is itself a carrier question.

2 Qubits from doublets, settings from repositioning

The CHSH configuration requires dichotomic observables with continuously tunable settings. Both are supplied natively.

The configuration. Carrier $\mathcal{C}_{641}$ ($641 = 4 \cdot 160 + 1$ prime, $640 = 2^7 \cdot 5$, primitive generator $g = 3$); an Object pair $\mathcal{O}_{17} \times \mathcal{O}_{17}$ contributing cycles $C_{16} = \langle g^{40} \rangle$, $g_{\mathcal{O}} = 601$, with the half-turn identity $g_{\mathcal{O}}^8 = -1$; Subjects $\mathcal{S}_{41}$ contributing cycles C_{40} , whose core with each Object is $C_8 = \gcd(40, 16)$, so the forced quotient has exactly *two* outcomes: the measurement is natively dichotomic. *All admissibility checks exact (`composite.py`).*

Proposition 2.1 (Channel conditioning yields qubits). *A pointer channel of the core- C_8 Subject projects onto windings in one class mod 8; applied to a C_{16} Object this leaves the two-winding doublet $\{s, s + 8\}$. Conditioning both wings of a synchronised orbit state leaves*

$$\Psi_{ab} = |a\rangle|-a\rangle + |a+8\rangle|-a-8\rangle, \quad (1)$$

a two-qubit maximally entangled state in winding space, prepared entirely by the Lüders update of [1].

Proposition 2.2 (The forced readout is σ_x). *For a doublet state $\psi = u^s + \lambda u^{s+8}$ with phase λ , the forced two-outcome readout of $\mathcal{S}_{41}$ gives, in the matched channel $r \equiv s \pmod{8}$,*

$$\Psi(0) = 8(1 + \lambda), \quad \Psi(1) = 8g_{\mathcal{O}}^8(1 - \lambda), \quad (2)$$

so the outcome weights are $w_{\pm} \propto |1 \pm \lambda|^2$: an exact σ_x measurement on the doublet qubit, with the sign supplied by the half-turn $g_{\mathcal{O}}^8 = -1$. Verified exactly in $\mathbb{F}_{641}$ for all 16 windings and all 80 settings (`composite.py`).

Definition 2.3 (Settings are relational repositionings). An analyser setting is a repositioning of the Subject against the Object along the carrier, $u \mapsto c^{-1}u$ with $c = g^j$. On a winding doublet it acts as the relative phase $c^{\mp 8}$, i.e. $\zeta_{80}^{\mp j}$: phase granularity $2\pi/80 = \pi/40$. The setting dial is physical — Subject–Object relative carrier phase, the same relational datum that the corpus’s decoherence metric reads [2] — and its resolution is set by the carrier, not by the subsystems.

3 The CHSH computation

Theorem 3.1 (Singlet law and exact Tsirelson saturation). *In the configuration above, with wings repositioned by j_a, j_b and read in matched channels, the joint weights are $w(\alpha, \beta) \propto |1 + (-1)^{\alpha+\beta} \zeta_{80}^{j_b - j_a}|^2$, hence the correlator obeys the singlet law*

$$E(j_a, j_b) = \cos\left(\frac{\pi(j_a - j_b)}{40}\right) \quad (3)$$

exactly, as an identity in $\mathbb{Z}[\zeta_{80}]$. The CHSH functional at the native settings $j_a, j_{a'} = 0, 20$ and $j_b, j_{b'} = 10, 70$ equals

$$S = 3 \cos \frac{\pi}{4} - \cos \frac{3\pi}{4} = 2(\zeta_8 + \zeta_8^{-1}) = 2\sqrt{2}, \quad S^2 = 8, \quad (4)$$

as a cyclotomic ring identity; and over the exhaustive sweep of all 80^3 admissible setting triples, $|S|$ never exceeds $2\sqrt{2}$. No-signalling holds exactly: each party's marginal is $1/2$ independently of the far setting; and the ledger-to-carrier reduction $\zeta_{80} \mapsto g^8$ commutes with every joint weight. All statements machine-verified (`composite.py`: C4–C8).

Proposition 3.2 (The quantum boundary from above). *No FRC composite configuration exceeds $2\sqrt{2}$. The ledger embeds in $\mathbb{C}$ with Hermitian pairings, observables are involutions on the embedded space, and Cirel'son's theorem [5] applies to every such model; the bound is inherited, not imposed. Together with Theorem 3.1, FRC sits exactly on the quantum boundary: strictly above the classical and epistemic-restriction bound 2 [3, 6], strictly below the algebraic bound 4.*

Experimental confrontation. Loophole-free Bell experiments [7, 8, 9] measure CHSH values consistent with the quantum prediction and inconsistent with $S \leq 2$. Theorem 3.1 is the statement that FRC reproduces this — not approximately, and not by tuning: the violation, its exact ceiling, and the singlet law all follow from subgroup arithmetic in $\mathbb{F}_{641}$. The result separates FRC quantitatively from the model classes nearest to it: epistemic-restriction theories [6] cap at 2 and are now excluded as the formalism's classification; superdeterministic substrates purchase violation by correlating settings with sources, where FRC's settings are free relational dials and the correlation is carried openly by the conserved orbit offset. The Kochen–Specker stance of the companion — values exist only as Subject-relative quotient phases — now carries quantitative weight: relational values *and* quantum correlation strength, simultaneously.

Remark 3.3 (Why saturation is exact). $2\sqrt{2}$ is irrational, but the ledger is not rational: it is cyclotomic, and $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$ is a ledger element. The finite formalism meets the Tsirelson bound head-on, with the irrationality absorbed by the quarter-turn's eighth-root refinement — the same arithmetic that makes amplitudes complex makes the quantum boundary exactly attainable. Measured CHSH values, in FRC as in the laboratory, are rational frequency estimates of a cyclotomic exact value.

4 Consequences and next steps

Closed. The composite gate exists and costs no new premise: Definitions 1.1–2.3 use only the drive, the forced bases, the Lüders update, and the carrier's own relational dials. The companion's open items (1) and (2) — the composite rule and the Tsirelson computation — are resolved, in the strongest available form (exact saturation). Twelve machine checks, exact arithmetic throughout.

Unblocked. *T4, the real-versus-complex network game* [10]: the machinery now suffices — two Bell pairs, entangled-basis readouts constructible as forced bases on doublet composites, and all required analyser angles available at $\pi/40$ granularity; execution requires importing the protocol’s exact score operator, and is the next computation. *T5, the gravitationally-induced-entanglement rate*: with composites defined, the rate of orbit-coherence formation between two locked clusters coupled through the substrate becomes a well-posed counting problem, and the corpus’s identification of gravity with phase synchronisation [2] predicts a nonzero rate — the forward prediction the programme should file before the first BMV-class data.

The independence frontier. Proposition 1.2 converts the tensor-independence axioms contested in [11] into carrier arithmetic: how independent two sources can be is measured by what they share beyond the universal core. The network-game computation will therefore not merely check a score; it will state FRC’s position in that dispute quantitatively.

Reproducibility

The script `composite.py` resides in the `validation/` folder of the companion draft and is slated for the suite at <https://github.com/gamayos/frc-quantum-numeric>: carrier admissibility, conserved offset and orbit length, the σ_x readout identity (16×80 cases), the singlet law over $\mathbb{Z}[\zeta_{80}]$ (all 80 settings), composite reduction commutation, exact no-signalling, the 80^3 CHSH sweep, and the $\mathbb{Z}[\zeta_8]$ saturation identities. Recorded output: twelve **PASS** lines and `composite: all checks passed`.

References

- [1] Y. Akhtman. *Quantum Observation in Finite Ring Cosmology: Wave Function as Phase, Amplitude as Subject-Relative Projection, Probability as Coincidence Count*. Companion draft, June 2026.
- [2] Y. Akhtman and E. Voether. *Gravitation as Phase Synchronisation on a Finite Substrate*. Preprint, 2026.
- [3] J. S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1(3):195–200, 1964.
- [4] J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt. Proposed experiment to test local hidden-variable theories. *Physical Review Letters*, 23(15):880–884, 1969.
- [5] B. S. Cirel’son. Quantum generalizations of Bell’s inequality. *Letters in Mathematical Physics*, 4(2):93–100, 1980.
- [6] R. W. Spekkens. Evidence for the epistemic view of quantum states: A toy theory. *Physical Review A*, 75:032110, 2007.
- [7] B. Hensen et al. Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature*, 526:682–686, 2015.
- [8] M. Giustina et al. Significant-loophole-free test of Bell’s theorem with entangled photons. *Physical Review Letters*, 115:250401, 2015.
- [9] L. K. Shalm et al. Strong loophole-free test of local realism. *Physical Review Letters*, 115:250402, 2015.

- [10] M.-O. Renou, D. Trillo, M. Weilenmann, T. P. Le, A. Tavakoli, N. Gisin, A. Acín, and M. Navascués. Quantum theory based on real numbers can be experimentally falsified. *Nature*, 600:625–629, 2021.
- [11] T. Hoffreumon and M. P. Woods. Quantum theory based on real numbers cannot be experimentally falsified. arXiv:2603.19208, 2026.